

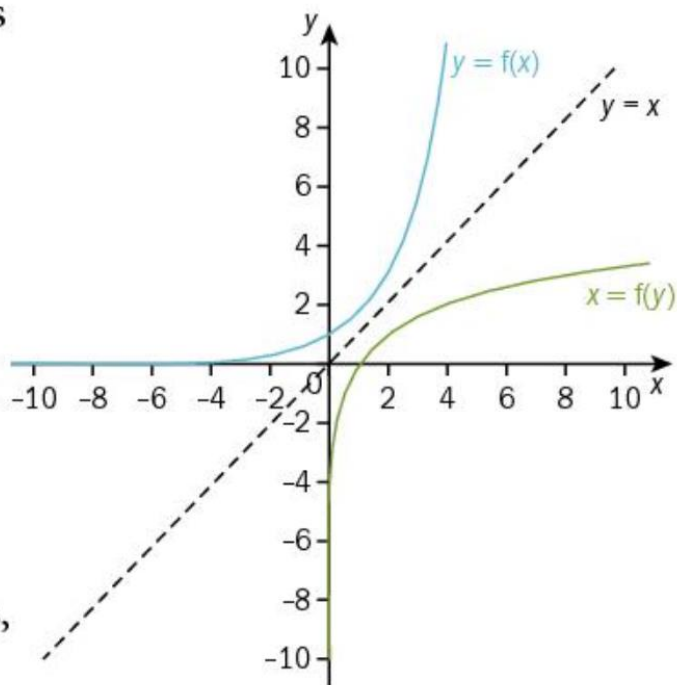
Logarithm Observatory — Logarithms and Exponentials

- a. Logarithmic function -> graph, inverse of exponential function

this

or

ers,



Example 5

Find the value of

- a) $\log_2 \frac{1}{8}$ b) $\log_8 2$ c) $\log_{\frac{1}{2}} 4$.

a) $\frac{1}{8} = 2^{-3} \Rightarrow \log_2 \frac{1}{8} = -3$

b) $2 = \sqrt[3]{8} = (8)^{\frac{1}{3}} \Rightarrow \log_8 2 = \frac{1}{3}$

c) $4 = \left(\frac{1}{2}\right)^{-2} \Rightarrow \log_{\frac{1}{2}} 4 = -2$

Example 6

a) Write these in the form $y = b^x$.

- i) $\log_2 64 = 6$ ii) $\log_k m = p$

b) Write these in the form $x = \log_b y$.

- i) $5^{-3} = 0.008$ ii) $h^5 = R$

a) i) $64 = 2^6$ ii) $m = k^p$

b) i) $\log_5 0.008 = -3$ ii) $\log_h R = 5$

- b. Law of logarithms

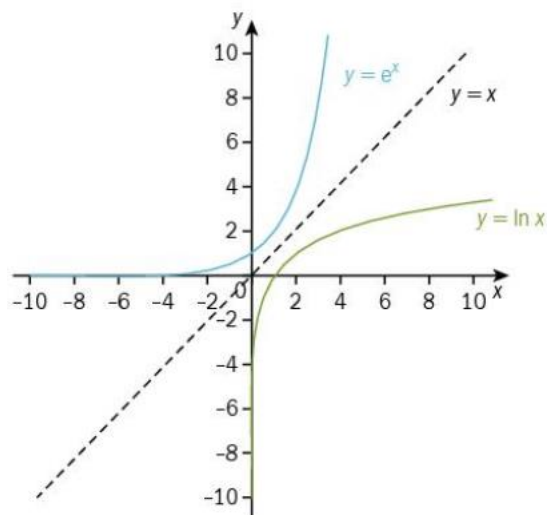
$$\log(xy) = \log x + \log y \quad (1)$$

$$\log\left(\frac{x}{y}\right) = \log x - \log y \quad (2)$$

$$\log(x^n) = n \log x \quad (3)$$

6. Express the following as a single logarithm or number.
- | | | |
|----------------------------|---|----------------------------------|
| a) $\log_5 3 + \log_5 8$ | b) $\log_6 7 + \log_6 10$ | c) $\log_5 4 + \log_5 6.25$ |
| d) $2\log_5 3 + \log_5 4$ | e) $2\log_5 6 - \log_5 8 + \log_5 12$ | f) $2\log_{10} 4 - 4\log_{10} 2$ |
| g) $2\log_p 4 - 4\log_p 2$ | h) $\log_{10} 12 - \frac{1}{2}\log_{10} 9 + \log_{10} 25$ | |
7. Express as a single logarithm.
- | | | |
|--------------------|----------------------|---|
| a) $\log_5 3 + 2$ | b) $3 + \log_6 10$ | c) $2\log_{10} x - \log_{10} 8 + \log_{10} (x + 3)$ |
| d) $3 + 2\log_5 x$ | e) $\log_{10} x + 3$ | f) $2 + \log_{10} 8 - \log_{10} x$ |
8. Express in terms of $\log x$, $\log y$, and $\log z$.
- | | | |
|--------------------------------------|------------------------------------|---|
| a) $\log\left(\frac{xy}{z^2}\right)$ | b) $\log\left(x^2(xy^2z)^3\right)$ | c) $\log\sqrt{\left(\frac{x^4y^2}{z^4}\right)}$ |
|--------------------------------------|------------------------------------|---|

c. E and log base e



$\ln x$ or $\log_e x$ is the inverse function of e^x .

d. Equations/inequalities with logs

Example 17

Solve the equation $\log_{10} (2 + x) = 2 + \log_{10} x$.

$$\log_{10} (2 + x) = 2 + \log_{10} x \Rightarrow \log_{10} (2 + x) = \log_{10} 100 + \log_{10} x$$

$$\Rightarrow \log_{10} (2 + x) = \log_{10} 100x \Rightarrow 2 + x = 100x \Rightarrow x = \frac{2}{99}$$

You need to express the right-hand side as a single logarithm.

Example 20

Solve the equation $\log_a (2-x) - 2\log_a x = \log_a 3$.

$$\begin{aligned}\log_a (2-x) - 2\log_a x &= \log_a 3 \Rightarrow \log_a \frac{(2-x)}{x^2} = \log_a 3 \\ \Rightarrow \frac{(2-x)}{x^2} &= 3 \Rightarrow 2-x = 3x^2 \Rightarrow 3x^2 + x - 2 = 0\end{aligned}$$

This is now a standard quadratic equation, with solutions $x = \frac{2}{3}, -1$.

However, only $x = \frac{2}{3}$ is valid because the logarithm of a negative number is undefined.

Having equated the expressions which were logarithms you need to check the 'solutions' actually gave a valid equation in log form.

Example 21

Solve the equation $\log_{10} (2-x) - 2\log_{10} x = 1$.

$$\begin{aligned}\log_{10} (2-x) - 2\log_{10} x &= 1 \Rightarrow \log_{10} \frac{(2-x)}{x^2} = \log_{10} 10 \\ \Rightarrow \frac{(2-x)}{x^2} &= 10 \Rightarrow 2-x = 10x^2 \Rightarrow 10x^2 + x - 2 = 0\end{aligned}$$

This is now a standard quadratic equation, with solutions $x = \frac{2}{5}, -\frac{1}{2}$ but only $x = \frac{2}{5}$ is valid.

You need to express both sides as a single logarithm in order to use this technique, so writing 1 as $\log_{10} 10$ is a critical step.

e. Equations/inequalities with exponentials

Example 15

Solve the equation $4^x = 16$ by taking logarithms of both sides.

$$4^x = 16 \Rightarrow x \log 4 = \log 16 \Rightarrow x = \frac{\log 16}{\log 4} = 2 \text{ (using any base)}$$

Example 16

Solve the equation $5^x = 3^{2x+1}$.

$$\begin{aligned}5^x &= 3^{2x+1} \Rightarrow x \log 5 = (2x+1) \log 3 \Rightarrow x(\log 5 - 2 \log 3) = \log 3 \\ \Rightarrow x &= \frac{\log 3}{\log 5 - 2 \log 3} = -1.869\end{aligned}$$

Since x appears in more than one power, you need to collect x terms together to solve for x .

Example 18

Solve the following inequalities.

a) $5^x \leq 13.3$ b) $(0.4)^x < 0.0001$

a) $5^x = 13.3 \Rightarrow x \log 5 = \log 13.3 \Rightarrow x = \frac{\log 13.3}{\log 5} = 1.608$
so $5^x \leq 13.3$ when $x \leq 1.608$

Powers of 5 give a monotonically increasing function, since $5 > 1$.

b) $(0.4)^x = 0.0001 \Rightarrow x \log 0.4 = \log 0.0001$
 $\Rightarrow x = \frac{\log 0.0001}{\log 0.4} = 10.1$
so $(0.4)^x < 0.0001$ when $x > 10.1$

Powers of 0.4 give a monotonically decreasing function, since $0.4 < 1$, so the direction of the inequality must be reversed.

f. Logs to reduce equations to linear

Exponential growth or decay:

$$y = ab^t \Rightarrow \log y = \log a + t \log b$$

The graph of 'log y' against 't' has intercept 'log a' and gradient 'log b'.

Power law:

$$y = ax^n \Rightarrow \log y = \log a + n \log x$$

The graph is of 'log y' against 'log x' and has intercept 'log a' and gradient n.

You know some examples of power laws in geometrical formulae for areas and volumes.

Example 23

By taking logarithms, transform these relationships between the two stated variables into a linear relationship between two new variables, and **state** the new variables.

- a) y and t are related by $y = 7b^t$.
- b) y and x are related by $y = ax^3$.

-
- a) $\log y = \log 7 + t \log b$ is linear with variables 'log y' and 't'.
 - b) $\log y = \log a + 3 \log x$ is linear with variables 'log y' and 'log x'.

